by providing sole carbon source. For screening microorganisms degrading the compound, different inhibitors are employed which have the ability to block a specific metabolic pathway of the non–target microbe, pH and temperature are also adjusted favoring the growth of desired microorganisms. Soil Calcium carbonate enrichment technique is used for isolation of secondary metabolite producing microorganisms (actinomycetes).

Secondary screening

It is very useful in sorting out microorganisms that have real commercial value from many isolates obtained during primary screening.

HOTS

Why microorganisms are exploited more than plant and animal cells for production of commercial products?

1. As primary screening allows the deflection and isolation of microorganisms which posses, potentially interesting industrial applications. It is further followed by secondary screening, to check the capabilities and gain information about these organisms.
2. Through primary screening only few or many microorganism that produce a industrially important product, are isolated. The information about the product that formed is very less. So, through secondary screening, further sorting out is performed. In this method, only microorganisms with real commercial value are selecting and those it lacks the potential are discarded.
3. Secondary screening should yield the types of information which are needed in order to evaluate the true potential of a micro organism industrially usage.
4. Secondary screening may be qualitative and quantitative in its approach.
5. It is done by using paper, thin layer or other chromatographic techniques.
6. The product's physical, clinical, and biological properties of the product are determined.
7. It detects gross genetic instability in microbial cultures.
8. It gives information about the number of products produced in a single fermentation.
9. It determines the optimum conditions for growth or accumulation of a product associated with particular culture.
10. It gives information about the different components of the fermentation medium.
11. It helps in providing information regarding the product yield potential of different isolates.
12. It reveals whether microorganisms are capable of a chemical change or of even destroying their fermentation product.

There are various methods employed for secondary screening which includes test conducting on petridish containing solid media or by using flasks or small fermentors containing liquid media, giant colony technique, and filtration method liquid medium method (using Erlenmeyer flask). Here giant colony technique's explained in detail.

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